

Univerzitet u Tuzli
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Telekomunikacione mreže

TK 210

Zadaća 1

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Broj indeksa: 22159

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1 Uvod

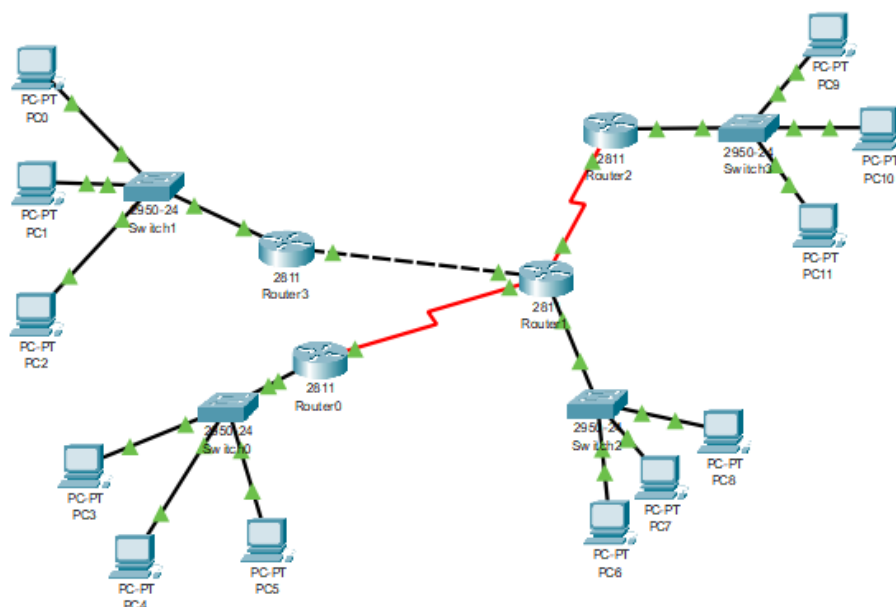
Cisco Packet Tracer je simulacijski alat razvijen od strane kompanije Cisco Systems s ciljem edukacije i praktičnog učenja mrežnih tehnologija. Ovaj softver omogućava korisnicima da kreiraju virtualna mrežna okruženja i testiraju konfiguracije bez potrebe za stvarnom opremom. U kontekstu predmeta Telekomunikacione mreže, Packet Tracer se koristi za razumijevanje rada mrežnih protokola, konfiguraciju rutera i svičeva te testiranje komunikacije između različitih mrežnih uređaja.

Program podržava razne mrežne protokole kao što su TCP/IP, RIP, DHCP, DNS i mnogi drugi, što omogućava korisnicima da steknu uvid u njihov stvarni rad. Omogućava i simulaciju rada lokalnih (LAN) i širokopojasnih (WAN) mreža, čime se povezuje teorijsko znanje sa praktičnim vještinama. Korištenjem ovog paketa doprinosi se razvoju sposobnosti rješavanja problema, logičkog razmišljanja i razumijevanja mrežne infrastrukture.

Osim konfiguracije mrežnih uređaja putem komandne linije (CLI), alat pruža i vizuelni prikaz toka podataka između čvorova, što dodatno pomaže u razumijevanju mrežnog saobraćaja. Korisnicima je također omogućeno eksperimentisanje sa raznim topologijama mreže bez ikakvog rizika. Kroz rad u Packet Traceru uče se osnovni i napredni koncepti mrežnog adresiranja, rutiranja, preklapanja (switching), te sigurnosti mreže.

Prednost ovog softvera je i mogućnost zajedničkog rada i učenja, jer podržava timsku saradnju i dijeljenje projekata. Učenje kroz simulaciju omogućava dublje razumijevanje složenih mrežnih scenarija koje bi bilo teško postaviti u stvarnim laboratorijskim uslovima.

2 Topologija mreže



Topologija telekomunikacijske mreže opisuje način na koji su uređaji i čvorovi međusobno povezani kako bi omogućili prenos podataka. Najčešće topologije su magistralna, zvjezdasta, prstenasta i mrežna, a svaka ima svoje prednosti u pogledu pouzdanosti i troškova. Pravilno odabrana topologija omogućava efikasan protok informacija, lakše održavanje i bolju otpornost na kvarove. U savremenim mrežama često se koristi hibridna topologija koja kombinuje više osnovnih struktura radi veće fleksibilnosti.

Na osnovu zadane topologije mi ćemo odraditi projekat koji slijedi u nastavku.

3 Naredbe za omogućavanje RIP protokola

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 192.168.21.1
Router(config-router)#network 192.168.21.218
Router(config-router)#no auto-summary
Router(config-router)#exit
Router(config)#exit
Router#
*SYS-5-CONFIG_I: Configured from console by console
```

```
Router>show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.21.1    YES manual up          up
FastEthernet0/1 unassigned      YES unset up          down
Serial0/1/0     192.168.21.218 YES manual up          up
Serial0/1/1     unassigned      YES unset down        down
Vlan1           unassigned      YES unset administratively down down
```

Figure 1: Nulti ruter

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.21.210
Router(config-router)#network 192.168.21.193
Router(config-router)#network 192.168.21.217
Router(config-router)#network 192.168.21.214
Router(config-router)#no auto-summary
Router(config-router)#exit
Router(config)#exit
Router#
*SYS-5-CONFIG_I: Configured from console by console
```

```
Router>show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.21.210  YES manual up          up
FastEthernet0/1 192.168.21.193 YES manual up          up
Serial0/3/0     192.168.21.217 YES manual up          up
Serial0/3/1     192.168.21.214 YES manual up          up
Vlan1           unassigned      YES unset administratively down down
Router>
```

Figure 2: Prvi ruter

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.20.1
Router(config-router)#network 192.168.21.213
Router(config-router)#no auto-summary
Router(config-router)#exit
Router(config)#exit
Router#
*SYS-5-CONFIG_I: Configured from console by console
```

```
Router>show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.20.1   YES manual up          up
FastEthernet0/1 unassigned      YES unset up          down
Serial0/3/0     192.168.21.213 YES manual up          up
Serial0/3/1     unassigned      YES unset down        down
Vlan1           unassigned      YES unset administratively down down
Router>
```

Figure 3: Drugi ruter

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.21.129
Router(config-router)#network 192.168.21.209
Router(config-router)#no auto-summary
Router(config-router)#exit
Router(config)#exit
Router#
*SYS-5-CONFIG_I: Configured from console by console
```

```
Router>show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.21.129 YES manual up          up
FastEthernet0/1 192.168.21.209 YES manual up          up
Serial0/3/0     unassigned      YES unset down        down
Serial0/3/1     unassigned      YES unset down        down
Vlan1           unassigned      YES unset administratively down down
Router>
```

Figure 4: Treći ruter

4 Dokazi uspješnog pinganja

```
C:\>ping 192.168.21.132

Pinging 192.168.21.132 with 32 bytes of data:

Reply from 192.168.21.132: bytes=32 time<1ms TTL=128
Reply from 192.168.21.132: bytes=32 time<1ms TTL=128
Reply from 192.168.21.132: bytes=32 time<1ms TTL=128
Reply from 192.168.21.132: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.21.132:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.21.131

Pinging 192.168.21.131 with 32 bytes of data:

Reply from 192.168.21.131: bytes=32 time<1ms TTL=128
Reply from 192.168.21.131: bytes=32 time<1ms TTL=128
Reply from 192.168.21.131: bytes=32 time<1ms TTL=128
Reply from 192.168.21.131: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.21.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 5: Pinganje 1

```
C:\>ping 192.168.21.2

Pinging 192.168.21.2 with 32 bytes of data:

Reply from 192.168.21.2: bytes=32 time=1ms TTL=125
Reply from 192.168.21.2: bytes=32 time=1ms TTL=125
Reply from 192.168.21.2: bytes=32 time=1ms TTL=125
Reply from 192.168.21.2: bytes=32 time=1ms TTL=125

Ping statistics for 192.168.21.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 192.168.21.3

Pinging 192.168.21.3 with 32 bytes of data:

Reply from 192.168.21.3: bytes=32 time=1ms TTL=125
Reply from 192.168.21.3: bytes=32 time=1ms TTL=125
Reply from 192.168.21.3: bytes=32 time=1ms TTL=125
Reply from 192.168.21.3: bytes=32 time=1ms TTL=125

Ping statistics for 192.168.21.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

Figure 6: Pinganje 2

```
C:\>ping 192.168.21.4

Pinging 192.168.21.4 with 32 bytes of data:

Reply from 192.168.21.4: bytes=32 time=1ms TTL=125
Reply from 192.168.21.4: bytes=32 time=1ms TTL=125
Reply from 192.168.21.4: bytes=32 time=1ms TTL=125
Reply from 192.168.21.4: bytes=32 time=1ms TTL=125

Ping statistics for 192.168.21.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 192.168.21.194

Pinging 192.168.21.194 with 32 bytes of data:

Reply from 192.168.21.194: bytes=32 time<1ms TTL=126
Reply from 192.168.21.194: bytes=32 time<1ms TTL=126
Reply from 192.168.21.194: bytes=32 time<1ms TTL=126
Reply from 192.168.21.194: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.21.194:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 7: Pinganje 3

5 Tabele dodijeljenih IP adresa

Uređaj	Interfejs	IP adresa	Maska	Povezan sa
Router1	FastEthernet0/0	192.168.21.210		R0, R3, R2, Switch2
Router1	FastEthernet0/1	192.168.21.193		R0, R3, R2, Switch2
Router1	Serial0/3/0	192.168.21.217		R0, R3, R2, Switch2
Router1	Serial0/3/1	192.168.21.214		R0, R3, R2, Switch2
PC0	FastEthernet0	192.168.21.130	255.255.255.192	Switch1
PC1	FastEthernet0	192.168.21.131	255.255.255.192	Switch1
PC2	FastEthernet0	192.168.21.132	255.255.255.192	Switch1
PC3	FastEthernet0	192.168.21.2	255.255.255.128	Switch0
PC4	FastEthernet0	192.168.21.3	255.255.255.128	Switch0
PC5	FastEthernet0	192.168.21.4	255.255.255.128	Switch0
PC6	FastEthernet0	192.168.21.194	255.255.255.194	Switch2
PC7	FastEthernet0	192.168.21.195	255.255.255.194	Switch2
PC8	FastEthernet0	192.168.21.196	255.255.255.194	Switch2
PC9	FastEthernet0	192.168.20.2	255.255.255.0	Switch3
PC10	FastEthernet0	192.168.20.3	255.255.255.0	Switch3
PC11	FastEthernet0	192.168.20.4	255.255.255.0	Switch3

Table 1: Tabela IP adresa između mrežnih uređaja

U ovoj tabeli se nalaze IP adrese, međutim nisu sve postavljene. Pronalazimo ih tako što kliknemo na neki PC (npr. PC0) te odaberemo opciju desktop pa IPconfiguration. Tu ćemo vidjeti IP, Subnet Masku. Za rutere odaberemo CLI ili CONFIG, pogledamo aktivne interfejsa i upišemo komandu: show ip interface brief.